Model Development Document

Stage-1 Regulatory Gate — CMPL-REG-24
EDA · 80/10/10 split · LogReg vs XGBoost vs LightGBM vs BERT
validation-optimized cut-off · OOV analysis · sensitivity analysis

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Generated: 2026-07-22 19:40 UTC
Data: data/complaints/cfpb_complaints.csv (4,000 rows)
Split: 5% scoring holdout, then 80/10/10 stratified, seed 42
Champion: lightgbm @ cut-off 0.883
Repo: reg-agents - model CMPL-REG-24 stage 1
Artifacts: docs/model_development/artifacts/
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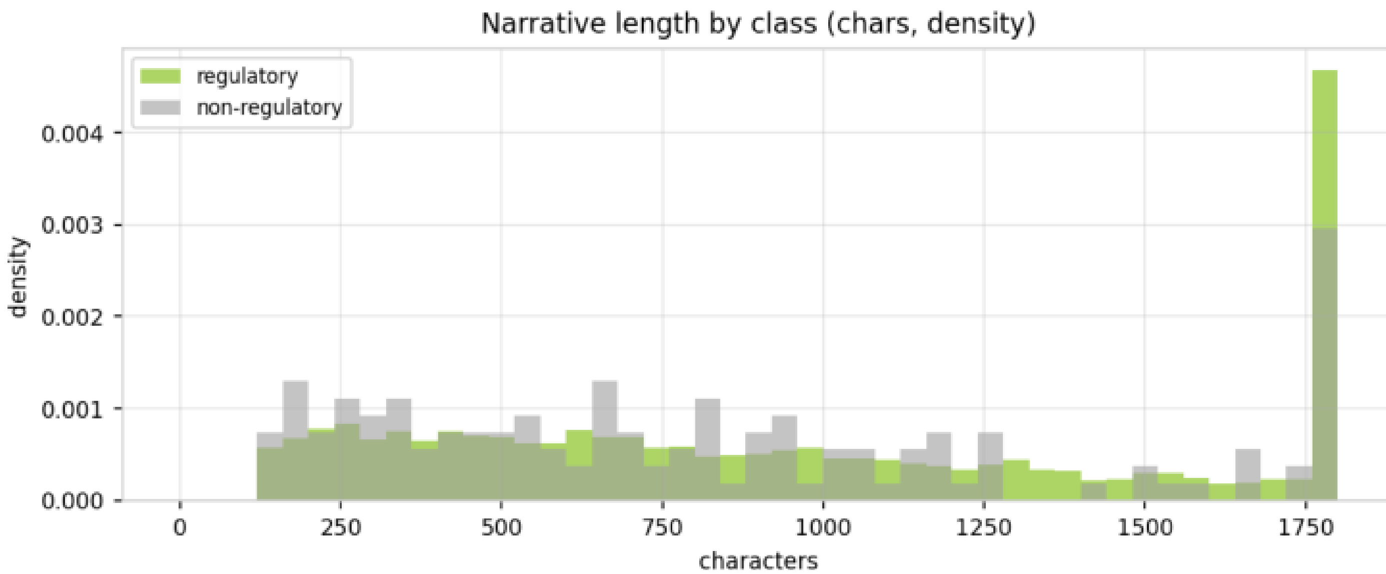
Abstract

This document records the development of the stage-1 regulatory gate for CMPL-REG-24 on 4,000 curated CFPB complaint narratives (96.6% regulatory, 3.4% non-regulatory minority). Four minority-balanced classifiers — logistic regression, XGBoost, LightGBM, and a fine-tuned DistilBERT — were compared under a stratified 80/10/10 train/validation/test protocol (applied after reserving a 5% scoring holdout for the batch-ingestion layer), with selection on validation minority-class PR-AUC, a decision cut-off optimized on the validation fold (0.883 for the champion, in place of the default 0.5), and one-shot test reporting. The champion, lightgbm, achieved validation minority PR-AUC 0.3011 and test minority PR-AUC 0.2467 (ROC-AUC 0.8459). Out-of-vocabulary and sensitivity analyses (threshold sweep, input perturbations, class-weight ablation, split-seed stability) bound the model's robustness. Reference labels are weak supervision from the CFPB taxonomy, and with only ~13 minority cases per held-out fold, minority metrics carry material split variance; both caveats condition the deployment recommendation.

2 · Data — class balance (Table 1) & properties (Table 2)

class / property	n / value	share
regulatory (1)	3,865	96.6%
non-regulatory (0)	135	3.4%
chars — reg / non-reg	870 / 715 (median)	
words — reg / non-reg	157 / 135 (median)	
'XXXX' PII masks per narrative	8.7 (mean)	

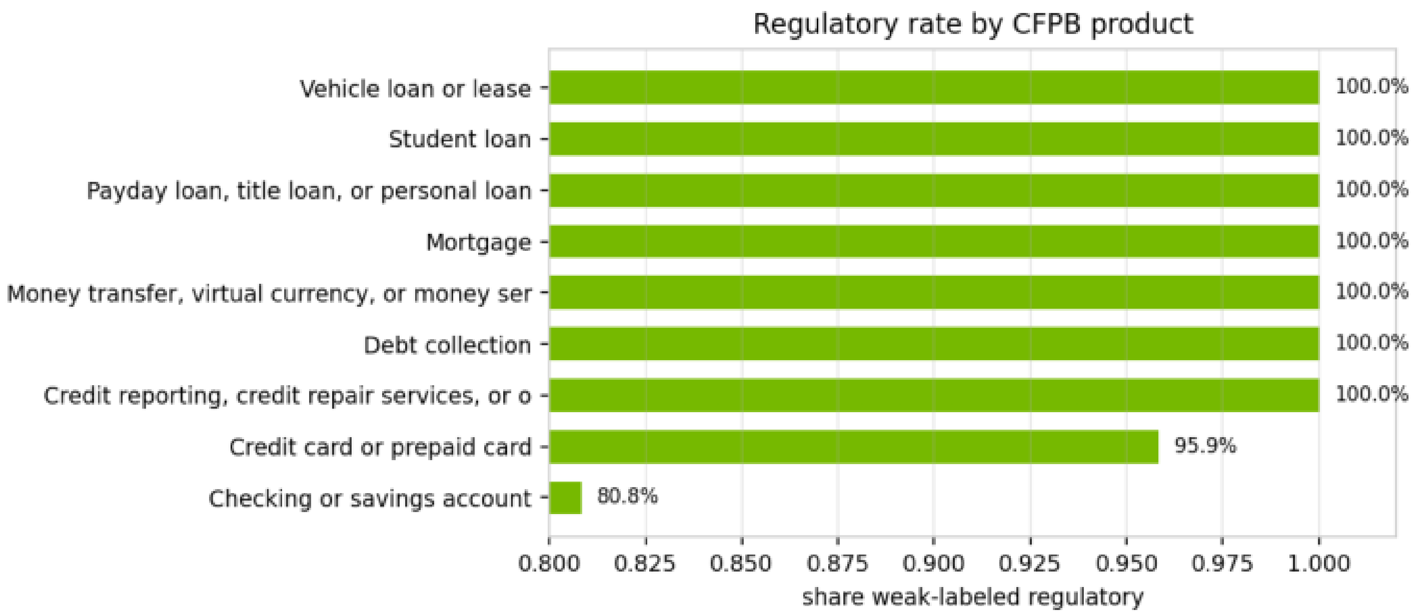
3 · EDA — narrative length by class (Fig. 1)



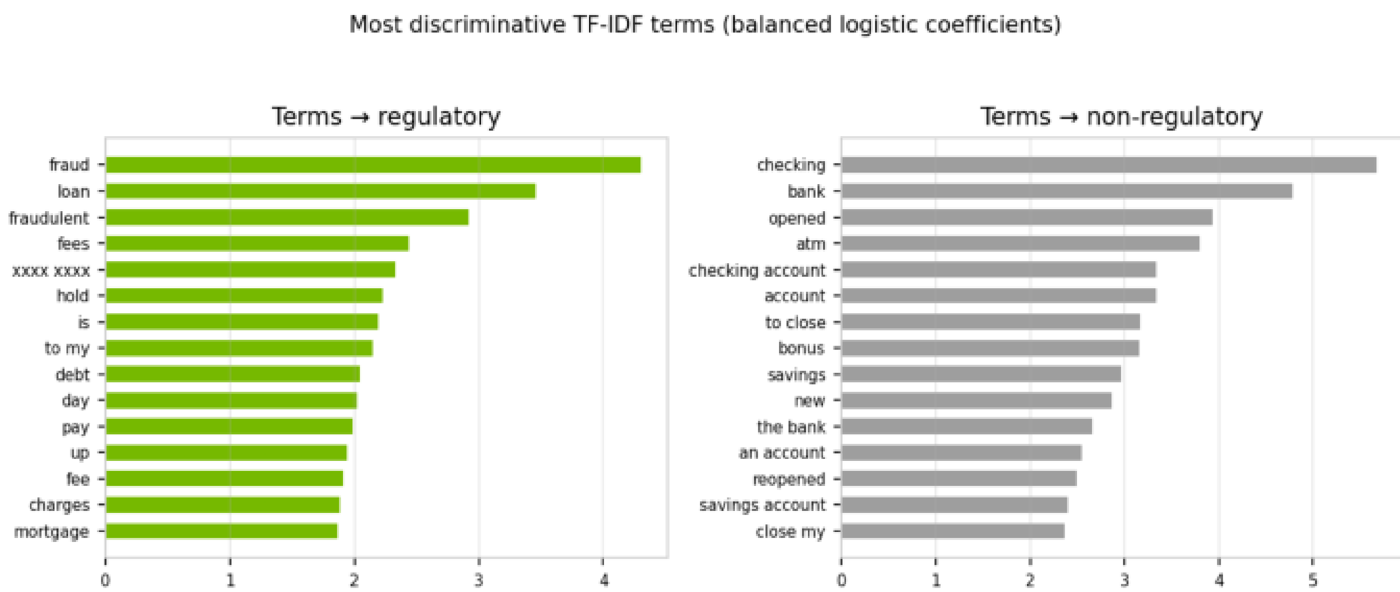
3 · EDA — regulatory rate by product (Table 3)

product	n	regulatory rate
Checking or savings account	506	80.8%
Credit card or prepaid card	916	95.9%
Credit reporting, credit repair servic	581	100.0%
Debt collection	597	100.0%
Money transfer, virtual currency, or	355	100.0%
Mortgage	364	100.0%
Payday loan, title loan, or personal	199	100.0%
Student loan	150	100.0%
Vehicle loan or lease	332	100.0%

3 · EDA — products (Fig. 2)



3 · EDA — discriminative terms (Fig. 3)



4 · Split — stratified 80/10/10, seed 42 (Table 5)

split	rows	regulatory	non-regulatory	reg rate
train (80%)	3,040	2,938	102	96.64%
validation (10%)	380	367	13	96.58%
test (10%)	380	367	13	96.58%
scoring holdout (5% c	200	193	7	96.50%

5 · Candidates and hyperparameters (Table 6)

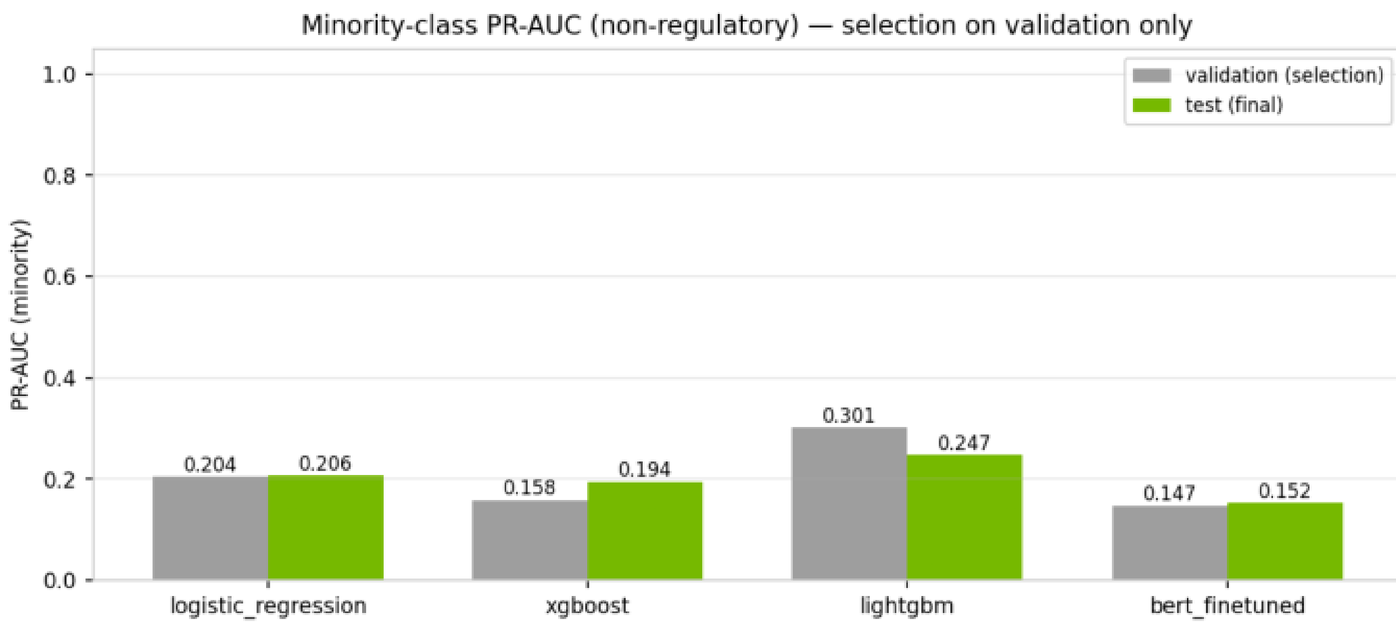
model	representation	hyperparameters	fit (s)
logistic_regression	TF-IDF 30k, 1-2 gram, subl	L2, C=4.0, max_iter=2000	0.05
xgboost	same TF-IDF features	300 trees, depth 6, lr 0.1, l =0.0	4.94
lightgbm	same TF-IDF features	400 trees, 63 leaves, lr 0.0	7.02
bert_finetuned	raw text → WordPiece subw	distilbert-base-uncased, m W lr 3e-5, 2 epo	98.1

6 · Leaderboard (Table 7)

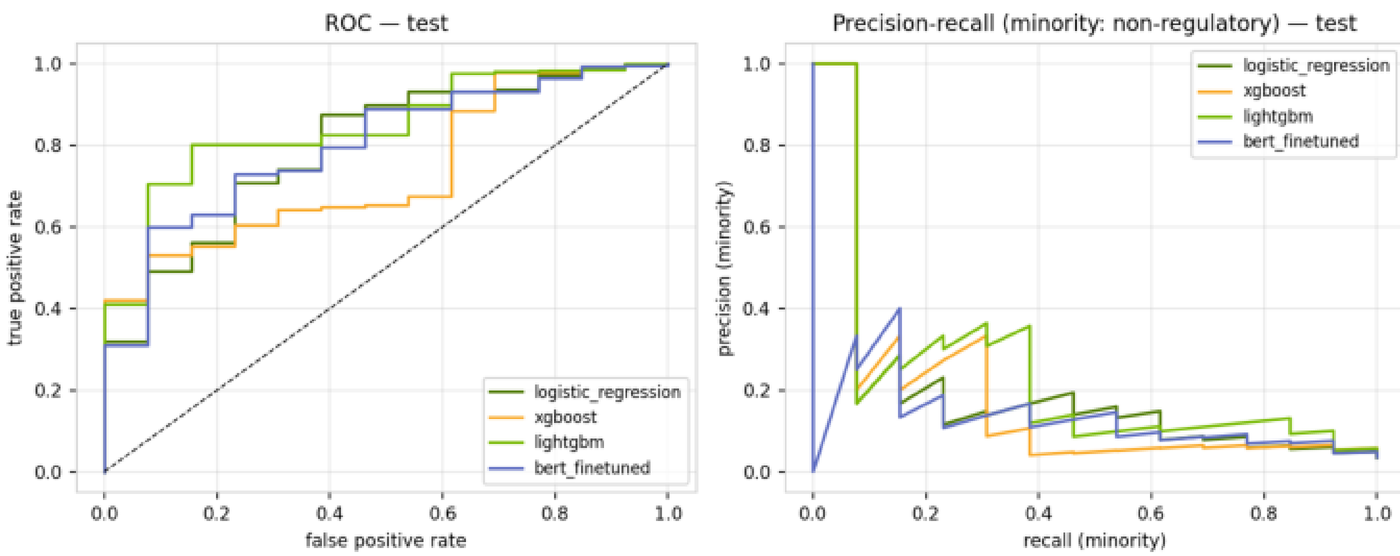
Champion: lightgbm - selected on validation minority PR-AUC; other columns are one-shot test metrics.

model	PR-AUC min (v	cut-off (val	PR-AUC min (te	ROC-AUC (tes	F1 min (test	bal-acc (test	Brier (test)	fit (s)
logistic_regr	0.204	0.788	0.206	0.7965	0.1538	0.584	0.0346	0.05
xgboost	0.1579	0.844	0.1938	0.7355	0.1702	0.613	0.0395	4.94
lightgbm	0.3011	0.883	0.2467	0.8459	0.1429	0.5385	0.0321	7.02
bert_finetur	0.1469	0.950	0.1517	0.7996	0.175	0.6875	0.0433	98.1

6 · Minority PR-AUC by model (Fig. 4)



6 · ROC and minority PR curves, test (Fig. 5)



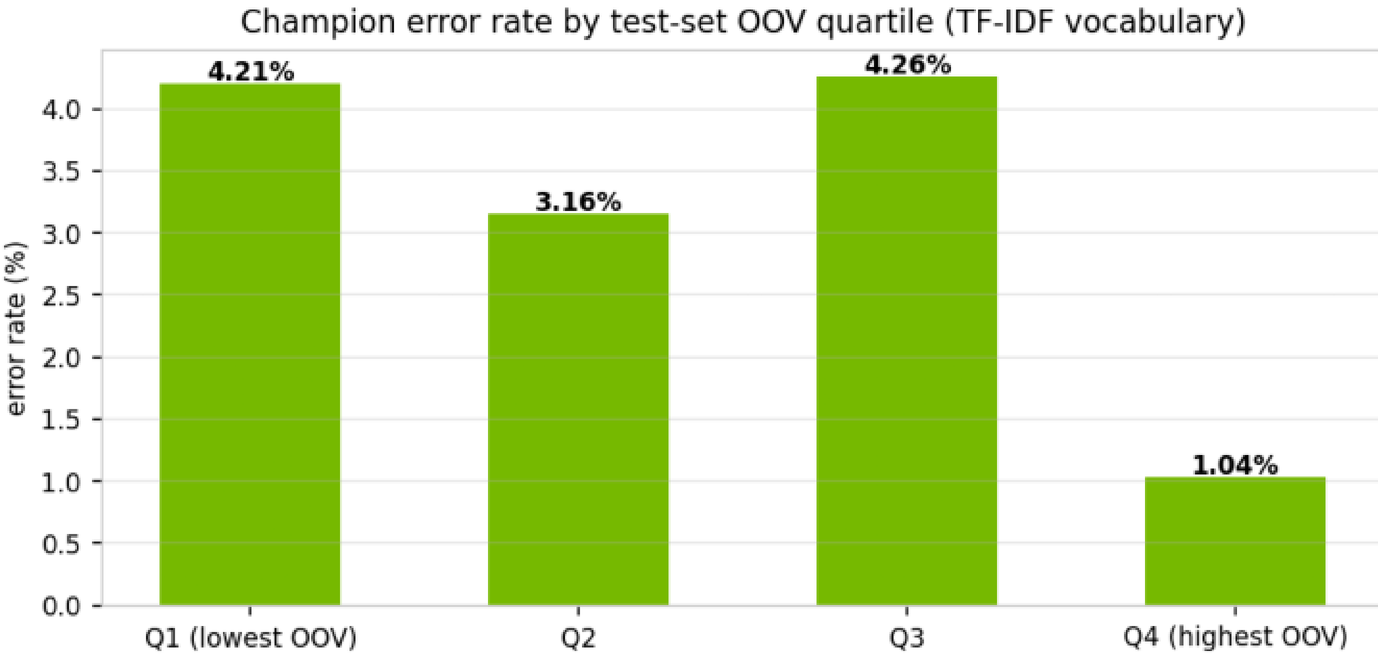
6 · Champion confusion matrix, test @ cut-off 0.883 (Table 8)

actual \ predicted	non-regulatory	regulatory
actual non-regulatory	1	12
actual regulatory	0	367

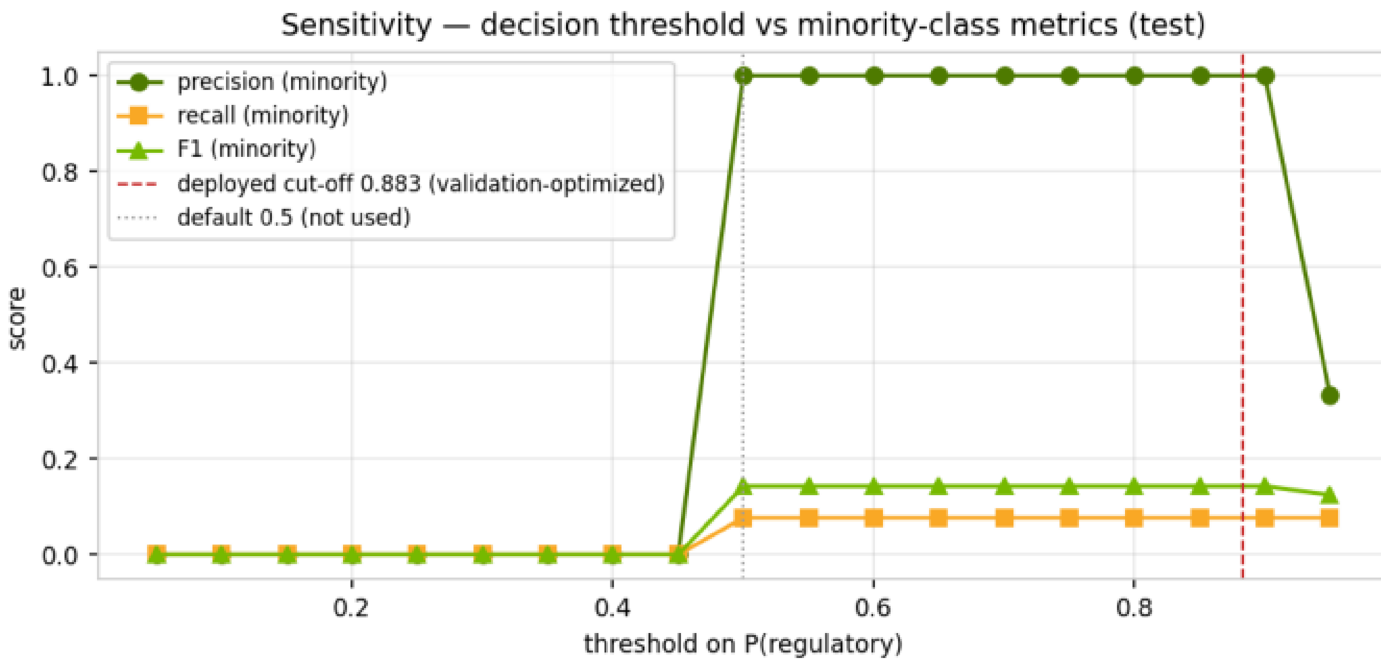
7 · OOV exposure (Tables 9-10)

OOV quartile / representatio	rate	n	error rate
TF-IDF vocabulary (30k, tra	2.0% of test tokens out-of-		
DistilBERT subword tokeniz	0.000% of test tokens map		
Q1 (lowest OOV)	0.1%	95	4.21%
Q2	1.0%	95	3.16%
Q3	2.1%	94	4.26%
Q4 (highest OOV)	4.6%	96	1.04%

7 · OOV analysis (Fig. 6)



8 · Threshold sensitivity (Fig. 7)



8 · Perturbation sensitivity (Table 11)

perturbation	PR-AUC (minority)	Δ
baseline (unperturbed)	0.2467	—
remove CFPB 'XXXX' PII masks	0.2792	+0.0325
lowercase + strip punctuation	0.2467	+0.0000
keep only first 50% of narrative	0.2332	-0.0135
randomly drop 10% of tokens	0.1399	-0.1068

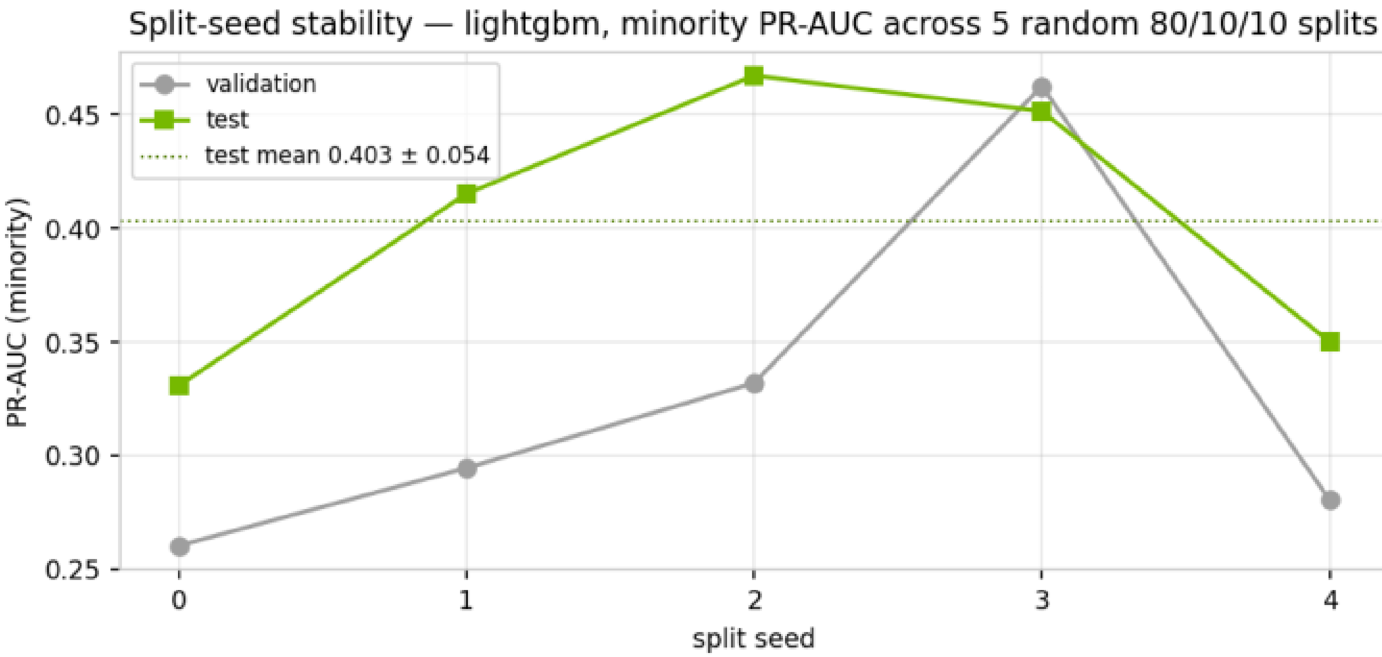
8 · Class-weight ablation (Table 12)

weighting	cut-off	PR-AUC min (val)	PR-AUC min (test)	F1 min (test)	bal-acc (test)
balanced (deploye	0.883	0.3011	0.2467	0.1429	0.5385
unweighted	0.982	0.2618	0.239	0.1176	0.5344

8 · Split-seed stability (Table 13)

seed	PR-AUC min (val)	PR-AUC min (test)
seed 0	0.2603	0.3307
seed 1	0.2945	0.4152
seed 2	0.3319	0.4672
seed 3	0.4626	0.4516
seed 4	0.2804	0.3500
mean ± std	0.3259 ± 0.0722	0.4029 ± 0.0542

8 · Seed stability (Fig. 8)



9 · Discussion and model selection decision

The bake-off selected lightgbm on validation minority PR-AUC. The spread between validation and test metrics, and the seed-stability band (0.3259 ± 0.0722 validation, 0.4029 ± 0.0542 test), shows that with only ~13 minority cases per held-out fold, single-split point estimates should not be over-read: the candidates are statistically close, and simplicity, latency, and interpretability are legitimate tie-breakers. The decision cut-off was optimized on the validation fold by maximizing minority F1 — the champion deploys at 0.883 rather than the default 0.5, which sits at a poorer point on the minority precision-recall trade-off (Figure 7). Perturbation analysis shows the model is robust to PII-mask removal and case/punctuation noise but degrades when narratives are truncated, so upstream text ingestion must preserve full narratives. OOV exposure is modest (2.0% of test tokens) and does not concentrate errors in the highest-OOV quartile. The class-weight ablation confirms minority balancing materially improves minority F1 at each variant's own optimized cut-off. The fine-tuned DistilBERT does not clear the bar to justify GPU serving cost for this gate today; it remains the upgrade path if complaint language drifts (subword tokenization eliminates OOV) or if golden-set labels reveal weak-label ceiling effects. Recommendation: lightgbm with the balanced weighting at the 0.883 cut-off is the research champion and promotion candidate; the deployed production gate keeps the champion and cut-off of the committed governance run (docs/complaint_model/metrics.json), with promotion via the standard change-management path. Revisit selection when the golden set lands or PSI drift alerts fire.

11 · Artifact manifest (Table 14)

artifact	size	description
tfidf_vectorizer.joblib	327 KB	TF-IDF featurizer (30k, 1-2 gram, sublinear, min_df
model_logistic_regression.joblib	223 KB	fitted candidate (challenger)
model_xgboost.joblib	77 KB	fitted candidate (challenger)
model_lightgbm.joblib	1,009 KB	fitted candidate (CHAMPION)
split_indices.json	41 KB	exact train/validation/test membership (reproducib
environment.txt	0 KB	package versions used for this run
(bert_finetuned weights)	—	fine-tuned distilbert-base-uncased (epochs=2, max

11 · Environment (Table 15)

package	version
python	3.11.15
numpy	1.26.4
pandas	2.2.3
scikit-learn	1.5.2
xgboost	2.1.1
lightgbm	4.7.0
torch	2.13.0
transformers	5.14.1
matplotlib	3.9.2

Sign-off

Prepared under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Independent review: submitted to second-line validation with this document, the data profile, and the artifact set under `docs/model_development/artifacts/`.